

Secant Method

This method is similar to Regula Falsi method. (1)

Working Rule:-

1) Let $f(x) = 0$ be the given equation.

2) Find x_0 and x_1 such that $f(x_0) < 0$ and $f(x_1) > 0$ (or $f(x_0)f(x_1) < 0$).

3) Find first approximate root by

$$x_2 = \frac{x_0 f(x_1) - x_1 f(x_0)}{f(x_1) - f(x_0)}$$

find $f(x_2)$

4) Find second approximate root by

$$x_3 = \frac{x_1 f(x_2) - x_2 f(x_1)}{f(x_2) - f(x_1)}$$

Calculate $f(x_3)$

5) Find third approximate root of (1) by

$$x_4 = \frac{x_2 f(x_3) - x_3 f(x_2)}{f(x_3) - f(x_2)}$$

Calculate $f(x_4)$

Repeat above process up to required approximate root.

$$x_{n+1} = \frac{x_{n-1} f(x_n) - x_n f(x_{n-1})}{f(x_n) - f(x_{n-1})}$$

where $n = 1, 2, 3, 4, \dots$

Qo:- find the real root of equation
 $x^3 - x - 1 = 0$ by Secant Method correct
 up to four decimal places.

Sol:- $f(x) = x^3 - x - 1$

To find x_0 and x_1 :-

$$f(0) = -1 < 0$$

$$f(1) = -1 < 0$$

$$f(2) = 8 - 2 - 1 = 5 > 0$$

$$f(1.5) = 0.875 > 0$$

$$f(1.4) = 0.343 > 0$$

$$f(1.3) = -0.103 < 0$$

choosing $x_0 = 1.3$ and $x_1 = 1.4$

$$f(x_0) = -0.103 \quad f(x_1) = 0.343$$

First approximation:-

$$x_2 = \frac{x_0 f(x_1) - x_1 f(x_0)}{f(x_1) - f(x_0)}$$

$$x_2 = 1.323042$$

$$f(x_2) = -0.007136 < 0$$

2nd approximation:-

$$x_3 = \frac{x_1 f(x_2) - x_2 f(x_1)}{f(x_2) - f(x_1)}$$

$$x_3 = 1.324605$$

$$f(x_3) = -0.000581 < 0$$

3rd approximation:-

$$x_4 = \frac{x_2 f(x_3) - x_3 f(x_2)}{f(x_3) - f(x_2)} = 1.324717$$

$$f(x_4) = -0.000009$$

$$x = 1.3247$$

Find a root of equation $x - e^{-x}$ correct to three decimal places using Secant Method. (2)

Sol:-

$$\text{Here } f(x) = x - e^{-x}$$

$$f(0) = -1$$

$$f(1) = 0.6321$$

$$f(0.5) = -0.1065$$

$$f(0.6) = 0.0512$$

$$\text{Let } x_0 = 0.5 \text{ and } x_1 = 0.6$$

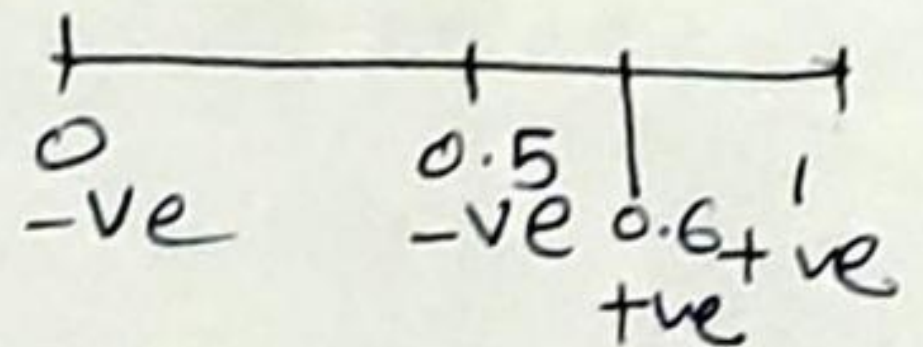
$$f(x_0) = f(0.5) = -0.1065,$$

$$f(x_1) = f(0.6) = 0.0512$$

By Secant Method

$$x_{n+1} = \frac{x_{n-1} f(x_n) - x_n f(x_{n-1})}{f(x_n) - f(x_{n-1})}$$

First approximation:-



$$x_2 = \frac{x_0 f(x_1) - x_1 f(x_0)}{f(x_1) - f(x_0)}$$

$$= \frac{0.5 (0.0512) - (0.6) (-0.1065)}{0.0512 - (-0.1065)}$$

$$x_2 = 0.5675$$

$$f(x_2) = 0.0006$$

Second approximation:-

$$x_3 = \frac{x_1 f(x_2) - x_2 f(x_1)}{f(x_2) - f(x_1)}$$

$$= \frac{(0.6)(0.0006) - (0.5675)(0.0512)}{0.0006 - 0.0512}$$

$$x_3 = 0.5671$$

$$f(x_3) = -0.00004$$

$$x_4 = \frac{x_2 f(x_3) - x_3 f(x_2)}{f(x_3) - f(x_2)}$$

$$x_4 = 0.5671$$

$$x_3 = x_4 = 0.567 \text{ (up to 3 decimal places)}$$

Hence the req. root is 0.567.

Qo:- Find the root of the equation $xe^x = \cos x$ using the secant method correct to four decimal places.